higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

INDUSTRIAL ELECTRONICS N6

22 November 2019

This marking guideline consists of 6 pages.

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QUESTION 1: TRANSIENTS

1.1
$$N = 1 + \frac{I_N 100}{S}$$
$$= 1 + \frac{I_N 100}{44,44} \quad \checkmark$$
$$= 1,104 \ \ oscillations \quad \checkmark$$
$$thus \ \ t = \frac{1}{f} \times N \quad \checkmark$$
$$= \frac{1 \times 1,104}{32,32} \quad \checkmark$$
$$= 34,158 \ \ ms$$
(5)

1.2
$$S = \frac{\pi R}{\omega L}$$
$$\therefore R = \frac{S(\omega L)}{\pi} \quad \checkmark$$
$$S = \frac{(2\pi fL)}{\pi} \quad \checkmark$$
$$R = S(2\ fL) \quad \checkmark$$
$$44,44\left[2 \times 32,32 \times 12 \times 10^{-3}\right] \quad \checkmark$$
$$= 34,471$$
$$= 34 \ \ \Omega \ \checkmark$$
(6)
[11]

QUESTION 2: TRANSDUCERS

2.1 Function: It decreases the voltage that is applied to the load.
Connection: R_D must be connected in series with the circuit load. (2)

2.2
$$V_I = \frac{I_L R_1 R_D}{R_2}$$
$$R_D = \frac{V_I R_2}{I_L R_1}$$
$$= \frac{1 \times 0,75 \times 10^3}{4 \times 10^{-3} \times 0,75 \times 10^3} \quad \vee \quad \frac{5 \times 0,75 \times 10^3}{20 \times 10^{-3} \times 0,75 \times 10^3}$$
$$= 250 \ \ \Omega$$
(3)

2.3 2.3.1 Current-to-voltage interfacing network (1)

2.3.2 a – Current source
b – R_1
c – R_2
d – R_D
e – Current load
f – R_1
g – Operational amplifier
h – R_2
i – R_L
j – V_O (10 × ½) (5)
[11]

QUESTION 3: ULTRASONICS

3.1 a – Power-supply unit
b – Ultrasonic generator
c – Ultrasonic transducer
d – Part to be cleaned/Jewellery
e – Cleaning fluid
f – Cleaning tub (6)

3.2
- Serves as a transport medium for the abrasive particles in the cutting fluid.
- Quenches any sparks that may exist.
- Keeps the workpiece cool.
- Washes away any burs which may form on the workpiece. (4)

[10]

QUESTION 4: X-RAYS AND RADIO ACTIVITY

4.1 $Dissipated\ \ power = P_T - Power\ \ used$

$P_T = Dissipated\ \ power + Power\ \ used$ ✓

$= 0,9 \times 10^3 + 6,774 \times 10^3$ ✓

$= 7\ \ 644\ \ W$

(3)

4.2 $X\text{-}ray\ \ power = P_T \times \eta$

$\eta = \frac{X\text{-}ray\ \ power}{P_T} = \frac{6,774 \times 10^3}{7\ \ 644} \times 100$

$= 88,619\,\%$ ✓ (4)

4.3 $P_{TOT} = V_{SUPPLY} \times I_{TOT}$

(3)

$$\therefore V_{SUPPLY} = \frac{P_{TOT}}{I_{TOT}} \quad \checkmark$$

$$¿ \frac{7\ 644}{8 \times 10^{-3}}$$

$$= 955{,}5\ \ kV$$

4.4 Physical magnitude:
The ionisation chamber is much bigger than the semiconductor detector.

Operating voltage:
The operating voltage required for the ionisation chamber is higher in value than the operating voltage required to operate a semiconductor detector. (2)
[12]

QUESTION 5: AUTOMATIC INSPECTION, TESTING AND NDT

5.1
5.1.1 D
5.1.2 E
5.1.3 A
5.1.4 B
5.1.5 C

(5 × 1) (5)

5.2
a – Search coil
b – Search coil oscillator
c – Mixer
d – Filter
e – Rectifier
f – Operational amplifier
g – Output device
h – Reference oscillator (8 × ½) (4)

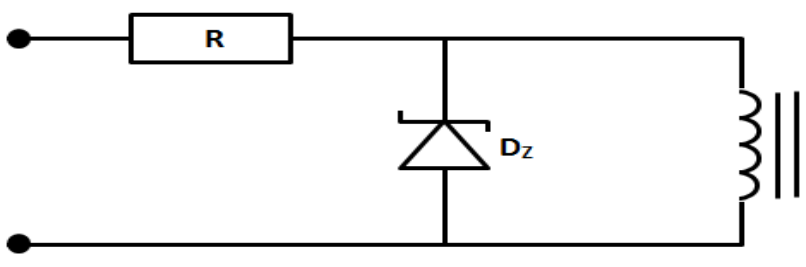
5.3 It allows for the detection of minute cracks and voids that are very small. (2)
[11]

QUESTION 6: ELECTRONIC SAFETY DEVICES

6.1
6.1.1 True
6.1.2 False
6.1.3 False
6.1.4 False
6.1.5 False

(5 × 1) (5)

6.2



(3)

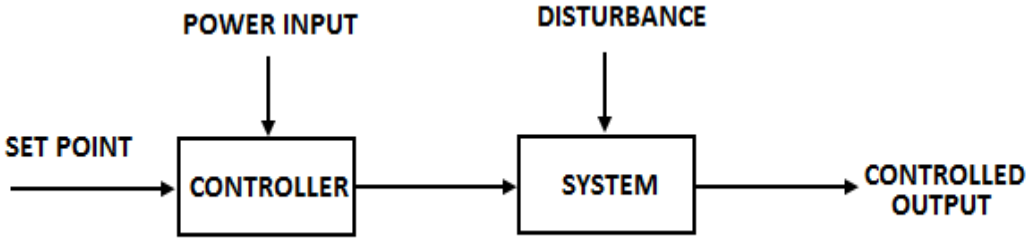
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6.3
- Phototransistors
- Photo diodes
- Photo conductors (3)

[11]

QUESTION 7: ELECTRONIC POWER CONTROL

7.1



(6)

7.2
- Regulator system
- Follow-up system (2)

7.3
- Task
- Design concept
- Computer
- Display (4)

[12]

QUESTION 8: THYRISTOR DEVICES AND SCR SPEED CONTROL

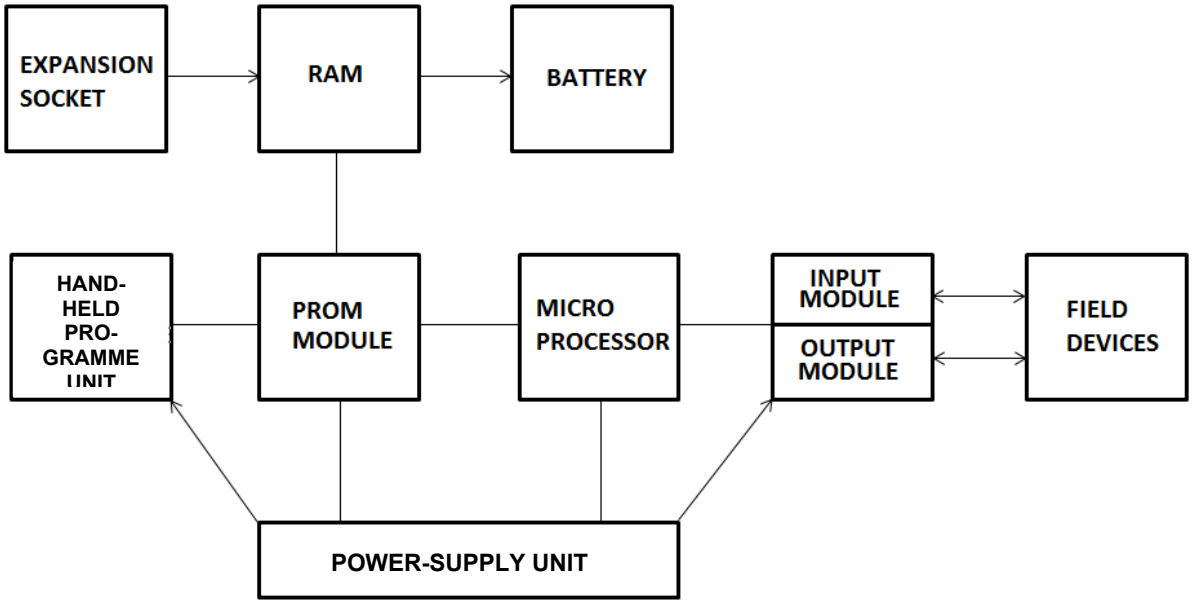
8.1
- Plugging
- Regenerative braking
- Dynamic braking (3)

8.2
- High efficiency
- Compact design
- Light weight
- High torque
- Low maintenance
- Wide range of designs
- Low motor cost (7)

8.3
- Constant voltage AC drives
- Constant current AC drives (2)

[12]

QUESTION 9: PROGRAMMABLE LOGIC CONTROLLERS



[10]

TOTAL: 100